Yukyeong Song

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EDUCATION

University of Florida **Gainesville, FL, USA**
Ph.D. in Curriculum and Instruction specializing in Educational Technology *Jan. 2022 – Aug. 2025*

- Advisor: Dr. Wanli Xing (Chair), Dr. Maya Israel (Co-chair)
- Cumulative GPA: 4.0/4.0
- Research Assistant (Full tuition & stipends): 2022 ~ 2025
- Doctoral Dissertation: *Designing a Generative AI-powered Teachable Agent to Engage All Middle School Students in Mathematics Learning*

Seoul National University **Seoul, Korea**
Master of Education in Educational Technology *Mar. 2019 – Feb. 2021*

- Advisor: Dr. Cheolil Lim
- Cumulative GPA: 4.16/4.3
- Lecture and Research Scholarship (Graduate Student Instructor, Full tuition & stipends): 2020
- Master's thesis: *Development of Data-Driven Debate (DDD) Instructional Model for Improving Data Literacy*

Sungkyunkwan University **Seoul, Korea**
Bachelor of Arts in Korean Language and Literature *Jan. 2014 – Feb. 2019*

- Cumulative GPA: 4.27/4.5
- Academic Excellence Scholarship: 2014-2018

University of North Carolina at Greensboro **Greensboro, NC, USA**
Exchange Student *Aug. 2016 – Jan. 2017*

RESEARCH INTERESTS

- AI in education
- STEM (e.g., AI, Computing, Data Science) education
- Learning analytics, Educational Data Science
- Fair, accountable, and transparent (FAccT) AI
- Universal design for learning

AWARDS & HONORS

1. New Student Organization of the Year Award, as President of Student Alliance of Graduates in Education (SAGE) *(Apr. 2025)*
2. Outstanding Student Award, Korean-American Scientists and Engineers Association (KSEA) (*Nov. 2024)*
3. Best Paper Award nomination, 2025 American Educational Research Association (AERA) conference (*Nov. 2024)*
4. Certificate of Outstanding Merit, International Center, University of Florida *(Nov. 2024)*
5. *Stephanie Kornprobst Wester Endowed Fellowship,* University of Florida, $4,000 *(Aug. 2024)*
6. Best Paper Award (Leading author), ACM Interaction Design and Children (IDC) conference (*June. 2023*)
7. Best Graduate Student Paper Award (Leading author), 2023 University of Florida College of Education's Research Symposium (*Apr. 2023*)
8. Best Paper Award (Leading author), ACM Learning Analytics and Knowledge (LAK) conference, Society for Learning Analytics Research (SoLAR) (*March. 2023*)
9. Best Paper Award, ACM Special Interest Group in Computer Science Education (SIGCSE) (*March. 2023*)

10. *Harold and Bette Riker Scholarship,* $1,500, University of Florida *(Aug. 2023)*
11. Outstanding Graduate Student in Research Award Nominee, University of Florida *(2023, 2024)*
12. Graduate Honorable Mention Poster, 2022 University of Florida College of Education's Research Symposium *(Apr. 2022)*
13. Golden Prize (1st place), Idea Festival (in-house idea contest) at Hyundai Motor Company $10,000 (*Dec. 2021*)
14. Exemplary researcher, Brain Korea 21 Graduate School Innovation Project, Seoul National University (*Jan. 2021*)

PROFESSIONAL WORK EXPERIENCE

University of Tennessee **Knoxville, TN, USA**
Tenure-track Assistant Professor *Aug. 2025– present*

University of Florida **Gainesville, FL, USA**
Graduate Research Assistant *Jan. 2022– Aug. 2025*

- Conducting research in multiple multimillion-dollar research projects.
- Design of emerging technologies in education.
- Curriculum design and implementation for AI/STEM/Computing education.

Hyundai Motor Company **Republic of Korea**
Manager, Research & Development Competency Development Team *Jan. 2021– Jan. 2022*

- Developed and implemented AI/Data Science online curricula for 12,000 car engineers.
- Launched an in-house AI contest that attracted around 400 engineers.
- Designed and implemented onboarding mentoring programs for more than 4,000 newly hired employees.
- Organized a series of online live book concerts for the Hyundai Motor Group.

Seoul National University **Republic of Korea**
Graduate Student Instructor (GSI), College of Education *March. 2020– Feb. 2021*

- Conducted and disseminated research, organized workshops, and events.
- Teaching assistant for undergraduate and graduate courses.
- Served at the College of Education administrative office.

the Academy of Creativity **Republic of Korea**
Editorial Secretary, International Journal of Creativity *Mar. 2019 – Feb. 2021*

- Assisted editors in manuscript review and managed the publication process.
- Organized monthly workshops and academic and practitioner events on creativity.

SNU Center for Innovation of Future Education, Seoul National University **Republic of Korea**
Research Assistant, Center for Innovation of Future Education *Mar. 2019 – Feb. 2021*

- Organized monthly software education workshops for undergraduate students and pre-service teachers.
- Conducted and disseminated research on emerging technologies and STEM education.

GRANTS

RESEARCH GRANT

***Funded* ($55,550)**

- *Improving Student Outcomes and Advance the Field of Individualized Instruction through AI Technology.* Successfully funded by New Schools Venture Fund, The Gates Foundation. ($*10,000*). PI: Dr. Chenglu Li, **Co-PI: Dr. Yukyeong Song**

- *Improving Student Outcomes and Advance the Field of Individualized Instruction through AI Technology.* Successfully funded by Accelerate: the National Collaborative for Accelerated Learning. ($*20,000*). PI: Dr. Chenglu Li, **Co-PI: Dr. Yukyeong Song**
- *AI Presidential Challenge Workshop Mentoring,* successfully funded by AI Tennessee Initiative, Office of Research, Innovation & Economic Development, University of Tennessee, Knoxville ($*15,550*). **PI: Dr. Yukyeong Song,** Co-PI: Dr. Rachel Wong. Nov. 2025 – May. 2026.
- *Elevating Higher Education with AI: CEHHS National Webinar Series and Open Course,* successfully funded by College of Education, Health, and Human Sciences Elevate Your Work Grant, University of Tennessee, Knoxville ($*10,000*). **PI: Dr. Yukyeong Song**, Co-PI: Dr. Chris Kilgore. Oct, 2025 – May. 2026.
- Contribution to writing the research grant proposal of *ALTER-Math: AI-augmented Learning by Teaching to Enhance and Renovate Math Learning*, successfully funded by Walton Family & Gates Foundations, Learning Engineering Virtual Institute (LEVI) program, # P0324564 *($10,000,000).* PI: Dr. Wanli Xing
- Contribution to writing a research grant proposal of *AI Across the Curriculum for Virtual Schools*, successfully funded by Department of Education, Education Innovation and Research (EIR) program (#S411C230070) ($*3,999,322*). PI: Dr. Jie Chao.
- Contribution to writing a research grant proposal of *Using Flow-Based Music Programming to Engage Children in Computer Science*, successfully funded by Innovative Technology Experiences for Students and Teachers (ITEST), National Science Foundation (NSF) (#2241715) ($*1,227,507*). PI: Dr. Wanli Xing.

Under review

- Submitted to National Science Foundation, CAMEL program: *Developing an AI-Ready Dataset of Authentic Human Tutoring Interactions in K-12 Mathematics Education*, PI: Zhikai Gao, Co-PI: **Dr. Yukyeong Song**
- Submitted to the Department of Education's FIPSE program (Artificial Intelligence area), PI: Zoi Apostolia Traga-Philippakos, Co-PI: Drs. Joshua Rosenberg, Emily Holtz, **Yukyeong Song**
- *Supporting Professional AI Readiness in K–12 (SPARK): Co-Designing Sustainable AI PD Through Research–Practice Partnerships in Rural East Tennessee,* submitted to National Science Foundation (NSF) K12 STEM Education, PI: Dr. Rachel Wong, Co-PIs: Drs. Lynn Hodge, **Yukyeong Song**, and Catherine Schuman.
- *Taking Adolescent Mental Health Interventions to the Next Level: Co-Creation of Mental Health Chatbots Powered by AI,* submitted to Ascent Program, Office of Research, Innovation, and Economic Development, University of Tennessee, Knoxville. PI: Dr. Kai Zhuang Shum, **Co-PI: Dr. Yukyeong Song**
- *REAL Science Lab,* submitted to Tools Competition, The Learning Agency. **PI: Dr. Yukyeong Song**, Co-PI: Dr. Chenglu Li
- Contribution to writing a scale-up grant proposal, *Project DIALOGS- scale up*, submitted to Innovative Technology Experiences for Students and Teachers (ITEST), National Science Foundation (NSF) ($*3,500,000*). PI: Dr. Maya Israel

Not Funded

- *CEHHS Graduate Research Assistant Funding Program*, submitted to the College of Education, Health, and Human Sciences, University of Tennessee, Knoxville. PI: Dr. Kai Zhuang Shum, **Co-PI: Dr. Yukyeong Song**
- Major contribution to writing a research grant proposal "DELTA-MATH: Diversified and Equitable Learning by Teaching AI for Math Learning" submitted to NSF Research on Innovative Technologies for Enhanced Learning (RITEL) program (*$900,000*). PI: Dr. Chenglu Li.

TRAVEL GRANTS ($ 16,730 in total)

- NSF-funded research projects conference travel grant (in total of $*15,480*)
- Graduate Student Council, University of Florida, conference travel grant (in total of $*550*)
- Graduate Studies Committee, School of Teaching and Learning, University of Florida conference travel grant (in total of $*700*)

RESEARCH EXPERIENCE

Research Assistant during Ph.D. Training

AI Across the Curriculum for Virtual Schools *Aug. 2024 – Aug. 2025*
(Funded by Department of Education EIR #S411C230070, $3,999,322; PI: Dr. Jie Chao (Concord), Co-PI: Dr. Wanli Xing)

- Design the AI-integrated math curriculum in virtual learning environments for high school students.
- Create the learning objectives and activities aligning with the AI big ideas and math standards.
- Design the user interface for virtual learning environments and conduct user testing.

AI Passport for Health *May. 2024 – Aug. 2025*
(Funded by University of Florida strategic funding)

- Designed and implemented a six-week curriculum to enhance AI literacy for physicians and medical students.
- Conducted research on proposing a framework for medical AI education.

ALTER-Math : AI-augmented Learning by Teaching to Enhance and Renovate Math Learning
(Funded by Walton Family & Gates Foundations, LEVI # P0324564, $10,000,000, PI: Dr. Wanli Xing) *Oct. 2023 – Aug. 2025*

- Technology interface and user experience design of the ALTER-Math teachable agent system.
- Conducted user studies with 140 students and a scale-up implementation for 1,500 middle school students.
- Learning analytics on students' language use and interactions log data with the system.

ART-Math: A Student-centered Interactive Mathematical Learning and Creation Platform powered by Artificial Intelligence
(Funded by Department of Education #91990023 C0022, $4,000,000, PI: Dr. Wanli Xing) *Apr. 2023 – Aug. 2025*

- Technology interface and user experience design on the ART-Math program.
- Generative AI interaction analysis with elementary school students in math learning.
- Led classroom studies involving 50 elementary school students and 20 teachers.

Collaborative Research: Innovating Quantum-Inspired Learning for Undergraduates in Research and Engineering (INQUIRE) *Oct. 2023 – Nov. 2024*
(Funded by NSF IUSE program #2142552, $1,250,000, PI: Dr. Gloria Kim, Co-PI: Dr. Guo Jing)

- Technology design of Spin-Qubit lab, a simulation-based learning technology.
- Form the theoretical foundations for the Spin-Qubit technology.
- Conducted the efficacy research with undergraduate and graduate students.

Fostering Computer Science and AI Learning through Youth-Led Conversational App Development Experiences
(Funded by NSF iTEST # DRL-2048480, $1,527,140; PI: Dr. Kristy Boyer, Co-PI: Dr. Maya Israel) *Jan. 2022 – Aug. 2024*

- Technology interface and user experience design of *AMBY* (AI Made By You),
  a conversational AI tinkering tool for racially minoritized middle school students (Black, African American students).
- Organized AI summer camp and professional development
  (230 middle school students and 20 undergraduates participated).

Using Flow-Based Music Programming to Engage Children in Computer Science
(Funded by NSF iTEST # 2241715, $1,227,507; PI: Dr. Wanli Xing, Co-PI: Dr. Maya Israel) *Jan. 2023 – Jan. 2024*

- Technology interface and user experience design of M-flow,
  a flow-based music programming tool for historically underserved elementary school students (Latinx students).
- Designed curriculum and conducted efficacy studies involving 79 elementary school students.

Precision Education: The Virtual Learning Lab
(Funded by Institute of Education Sciences: #R305C160004, $8,908,288; PI: Dr. Walter Leite) *Jan. 2022 – Jan. 2024*

- Conducted data-mining and learning analytics research with the collected log and textual data.
- Conducted FAccT AI research on the collected data.

Development of Instructional Methodology for Creative Thinking

(Funded by LG Academy, PI: Dr. Cheolil Lim) *Jul. 2019 – Nov. 2019*

- Project management
- Designed and implemented educational programs to enhance more than 1,000 LG employees' creative thinking.

Development of Creative Marketing Training Program
(Funded by CJ Foodville, PI: Dr. Cheolil Lim) *Jul. 2019 – Nov. 2019*

- Project management, leader
- Designed and implemented educational programs for CJ Foodville marketers.

PUBLICATIONS

ARTICLES PUBLISHED IN REFEREED JOURNALS

1. Zhu, W., Guo, R., Zhu, G., Li, C., & **Song, Y.** (2025). GAI4DE: Harnessing the design process to integrate GAI into design studies. *International Journal of Human-Computer Interaction.* DOI: https://doi.org/10.1080/10447318.2025.2603662 (IF: 5.5).
2. Xing, W., Kim, T., **Song, Y.,** Li, H., Li, C., & Kim, J. (2025). Unveiling interaction patterns between students and generative AI teachable agent: focusing on students' agency and AI agents' authority. *British Journal of Educational Technology.* DOI: https://doi.org/10.1111/bjet.70038 (IF: 7.2).
3. **Song, Y.,** Li, C., Ma, Y., Lyu, B., Zhu, W., Li, H., & Xing, W. (2025). Exploring effective tutoring strategies in asynchronous online mathematical discussions: insights from ordered network analysis. *Journal of Science Education and Technology*. DOI: https://doi.org/10.1007/s10956-025-10233-0
4. **Song, Y.,** Li, C., Xing, W., Lyu, B., & Zhu, W. (2025). Investigating perceived fairness of AI prediction system for math learning: A mixed-methods study with college students. *The Internet and Higher Education*, 101000. DOI: https://doi.org/10.1016/j.iheduc.2025.101000 (IF: 6.4).
5. Xing, W., **Song, Y.,** Li, C., Liu, Z., Zhu, W., & Oh, H. (2025). Development of a generative AI-powered teachable agent for middle school mathematics learning: a design-based research study. *British Journal of Educational Technology (BJET*). DOI: https://doi.org/10.1111/bjet.13586 (IF: 7.2).
6. **Song, Y**., Kim, J., Xing, W., Liu, Z., Li, C., & Oh, H. (2025). Elementary school students' and teachers' perceptions toward creative mathematical writing with Generative AI. *Journal of Research on Technology in Education,* 1-23. DOI: https://doi.org/10.1080/15391523.2025.2455057 (IF: 4.5).
7. **Song, Y**., Kim, J., Liu, Z., Li, C., Xing, W. (2025). Students' perceived roles, opportunities, and challenges of an AI-powered teachable agent in middle school math class. *Journal of Research on Technology in Education,* DOI: http://dx.doi.org/10.1080/15391523.2024.2447727 (IF: 4.5).
8. Lyu, B., Li, C., Li, H., Oh, H., **Song, Y**., Zhu, W., & Xing, W. (2025). The Role of Teachable Agents' Personality Traits on Student-AI Interactions and Math Learning. *Computers and Education*. (IF: 12).
9. Kim, J, Detrick, R., Yu, S., **Song, Y.,** Bol, L., Li, N. (2025). Socially shared regulation of learning and Generative AI: Opportunities to support socially shared regulation. *Education and Information Technologies.* DOI: 10.1007/s10639-024-13187-9 (IF: 4.8).
10. Li, C., Xing, W., **Song, Y.,** & Lyu, B. (2025). RICE AlgebraBot: lessons learned from designing and developing responsible conversational AI using induction, concretization, and exemplification to support algebra learning. *Computers and Education: Artificial Intelligence.* DOI: https://doi.org/10.1016/j.caeai.2024.100338.
11. **Song, Y.,** Xing, W., Li, C., Tian, X., Ma, Y. (2024). The relationship between math literacy and linguistic synchrony in online mathematical discussions. *British Journal of Educational Technology*. DOI: https://doi.org/10.1111/bjet.13444 (IF: 7.2).
12. **Song, Y.,** Weisberg, L., Zhang, S., Tian, X., Boyer, K. E., Israel, M. (2024). A framework for inclusive AI

learning design for diverse learners. *Computers and Education: Artificial Intelligence*, 100212. DOI: https://doi.org/10.1016/j.caeai.2024.100212.

13. Xing, W., Li, H., Kim, T., Zhu, W., & **Song, Y**. (2024). Investigating the behaviors of core and periphery students in an asynchronous online discussion community using network analysis and topic modeling. *Education and Information Technologies*, 1-28. DOI: https://doi.org/10.1007/s10639-024-13038-7 (IF: 4.8).

14. Rahimi, S., Almond, R., Ramíres-Salgado, A., Wusylko, C., Weisberg, L., **Song, Y.,** Lu, J., Myers, T., Wang, B., Wang, X., Francois, M., Moses, J., & Wright, E. (2024). Competency model development: the backbone of Successful Stealth Assessments. *Journal of Computer Assisted Learning.* DOI: https://doi.org/10.1111/jcal.13025 (IF: 5.4)

15. Tian, X., Kumar, A., Solomon, C., Calder, K., Katuka, G. A., **Song, Y**., Celepkolu, M., Pezzullo, L., Barrett, J., Boyer, K. E., and Israel, M. (2023). AMBY: A development environment for youth to create conversational agents. *International Journal of Child-Computer Interaction 38*, 100618, 1-15. DOI: https://doi.org/10.1016/j.ijcci.2023.100618 (IF: 5.4).

16. Lim, C., Lee, J., **Song, Y.,** Lee, U., Lee, H., & Lim, E. (2022). A study on the development and application of a data science education program for elementary school students (Korean). *The Journal of Educational Information and Media, 28*(2), 319-345. (KCI IF: 4.73)

17. **Song, Y**., Song, S., Kim, Y., & Lim, C. (2021). A developmental study of an instructional model and strategies for data- driven debate (DDD) to improve data literacy (Korean). *Journal of Educational Technology, 37*(4), 943-982. (KCI IF: 4.24)

18. **Song, Y.** & Lim, C. (2021). Development of an instructional model for design thinking-based AI service planning program for high school students (Korean). *The Journal of Educational Information and Media, 27*(2), 371-402.

19. Lim, C., **Song, Y.,** Hong, S., & Park, C. (2020). A Study on the application and improvement of the rapid prototyping to instructional systems design (RPISD) model (Korean). *Journal of Educational Technology, 36*(3), 589-617.

20. Lim, C., Han, H., Hong, S., **Song, Y.,** & Lee, D. (2020). Design principles for improving creative thinking competency in corporate education (Korean). *The Journal of Educational Information and Media, 26*(3), 477-510.

MANUSCRIPTS SUBMITTED FOR PUBLICATION

1. **Song, Y.,** Tian, X., Dobar, E., Wise, C., McKlin, T., Boyer, K. E., Israel, M. (under review – major revision submitted). A four-pronged approach to assessing artificial intelligence learning in a middle school summer camp. Manuscript submitted for publication to *ACM Transactions on Computing Education (TOCE).*

2. **Song, Y.,** Li, C., Xing, W., Li, S., Lee, H. H., Tan, Y., Shaffer, D. W., Ma, Y. (under review). Using fair AI to uncover the behavioural patterns of self-regulated learning in a virtual learning environment. Manuscript submitted for publication to *Educational Technology & Society (ETS*).

3. Zhang-Kopf, Y., Li, C., Gulfidan, G., Li, H., **Song, Y.**, Guo, R., Xing, W. (2026). When to stop? An experimental study on AI teachable agent stopping mechanisms and their learning affordances in mathematics. Manuscript submitted for publication in *Computers in Human Behavior.*

4. Xing, W., **Song, Y.,** Du, H., Zhu, W., Li, C. (under review- major revision submitted). Investigating expert and peer tutoring behaviors in a large online discussion forum using temporal dynamic analytics. Manuscript submitted for publication to the *Journal of Learning Analytics.*

5. Ma, Y., **Song, Y.,** Mehmet, C., Boyer, K. E. (under review). Automatically detecting confusion and conflict during collaborative learning using linguistic, prosodic, and facial cues. Manuscript submitted for publication to *User Modeling and User-Adapted Interaction.*

6. Ma, Y., **Song, Y**., Balch, J., Hu, Z., Kolla, S., Vellanki, D., Guan, Z., Ren, Y., Armfield, B., Ozrazgat-Baslanti, T. et al., (under review). Promoting AI competencies for medical students: a systematic review on frameworks, programs, and tools. Manuscript submitted for publication to *New England Journal of Medicine AI.*

7. Xing, W., Zhu, W., **Song, Y.,** Li, C., Lyu, B., Li, H., Liu, Z., & Zhang, F. (under review). Reading, solving, creating: a generative AI-powered math story creation system. Manuscript submitted for publication to *Computers and Education.*

8. Xing, W., Liu, Z., **Song, Y.** (under review). Why do students leave instructional videos: understanding students' in-video dropout behavior in a large online math learning platform? Manuscript submitted for publication to *Educational Technology Research and Development.*

9. Kim, J., Yu, S., **Song, Y**., Brown, E., & Mutiu, A. (under review). AI in the multigrade rural classroom: Exploring Nigerian teachers' perceived challenges and the push for the implementation of AI in education based on activity theory. Manuscript submitted for publication to *Education and Information Technologies*.

10. Kim, J., Yang, G., Chan, S. W., Lin, X, **Song, Y.,** & Li, N. (under review). Game-based learning for asynchronous online AI literacy course: Approach to improve students' cognitive, behavioral, and affective and ethical learning of AI. Manuscript submitted for publication to *International Journal of Educational Technology in Higher Education*.

11. Tian, X., **Song, Y.,** Zhang, S., M, Israel., Boyer, K. (under review). "My bot can talk about science!": Integrating AI into middle school science classrooms through chatbot development. Abstract submitted for publication to *Computers and Education.*

12. Tian, X., Zhang, S., **Song, Y.,** Mannekote, A., Barrett, J., Wise, C. F., Dobar, E., Boyer, K. E., & Israel, M. (under review). Too skeptical or not skeptical enough? Middle school student and teacher perceptions of LLM-based project assessment. Manuscript submitted for publication to *Education and Information Technologies.*

13. Go, B., Karademir, N., **Song, Y**., Lee, K. (in progress). Is artificial intelligence "intelligent" for education? A critical review of AIED research for who, how, and why.

PAPERS PUBLISHED IN REFEREED CONFERENCE PROCEEDINGS

1. Ma, Y., & **Song, Y.** (2026). Mapping AI Literacy in Medical Education: A Review of Concepts and Teaching Practices. In *Proceedings of The Fourteenth IEEE International Conference on Healthcare Informatics ICHI2026.* IEEE.

2. Ma, Y., & **Song, Y.** (2026). Early design of a case-based platform for training responsible AI practice in clinical care. In *Proceedings of The Fourteenth IEEE International Conference on Healthcare Informatics ICHI2026.* IEEE.

3. Zhang-Kopf, Y., Li, C., Gulfidan, G., Li, H., **Song, Y.**, Guo, R., Xing, W. (2026). When to stop? An experimental study on AI teachable agent stopping mechanisms and their learning affordances in mathematics. In *Proceedings of AI4EDU@AAAI 2026* (Communications in Computer and Information Science). Springer.

4. Kim, J., Detrick, R., **Song, Y.,** & Brown, E. (2025). Navigating AI in rural multigrade classrooms: Nigerian teachers' perspectives through the lens of activity theory. In *Proceedings of 2025 International Conference of the Learning Sciences* (ICLS 2025). DOI:

5. Tian, X., Zhang, S., **Song, Y**., McKlin, T., Boyer, K., & Israel, M. (2025). An attitude paradox? How a middle school conversational AI learning experience increased ability beliefs but lowered intention to persist. In *Proceedings of 2025 International Conference of the Learning Sciences* (ICLS 2025). DOI:

6. Lyu, B., Li, C., Li, H., Oh, H., **Song, Y**., Zhu, W., Xing, W. (2025). Exploring the role of teachable AI agents' personality traits in shaping student interaction and learning in mathematics education. In *Proceedings of 2025 International Conference on Learning Analytics & Knowledge* (LAK 2025)*,* March 3-7, Dublin, Ireland*.*

7. Tadimalla, Y., Figard, R., **Song, Y**. (2024). WIP: Moving from accessibility to anti-ableism through the explication of disability in the AI ecosystem. *In Proceedings of 2024 IEEE Frontiers in Education*, Oct 13–16, 2024, Washington, D.C., USA.

8. Liu, Z., Guo, R, **Song, Y**., Xing, W. (2024). WIP: understanding students' in-video dropout behavior in large online math learning platform. *In Proceedings of 2024 IEEE Frontiers in Education*, Oct 13–16, 2024, Washington, D.C., USA.

9. Oh, H., & Guo, R., & Xing, W., & Liu, Z., **Song, Y**., & Li, C. (2024, June). *The Seamless Integration of Machine Learning Education into High School Mathematics Classrooms*. Paper presented at the 2024 ASEE Annual Conference & Exposition, Portland, Oregon. DOI: 10.18260/1-2—48137

10. Tian, X., Mannekote, A., Solomon, C. E., **Song, Y**., Wise, C. F., Mcklin, T., Barrett, J., Boyer, K. E., & Israel, M. (2024). Examining LLM prompting strategies for automatic evaluation of learner-created computational artifacts. *In Proceedings of the 17th International Conference on Educational Data Mining (EDM)*, 1–4. DOI: 10.5281/zenodo.12729922.

11. Lee, H. H., Celedón-Pattichis, S., LópezLeiva, C. A., Pattichis, M. S., & **Song, Y.** (2024). Emergent interests to well-developed Interests in STEM. *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024*. International Society of the Learning Sciences. DOI: https://doi.org/10.22318/icls2024.111198.

12. **Song, Y**., Li, C., Xing, W., Li, S., & Lee, H. H. (2024). A fair clustering approach to self-regulated learning behaviors in a virtual learning environment. In *Proceedings of 2024 International Conference on Learning Analytics & Knowledge* (LAK 2024), March 18-22, Kyoto, Japan, p. 771-778. DOI: https://doi.org/10.1145/3636555.3636863.

13. **Song, Y**., Tian, X., Regatti, N., Katuka, G. A., Boyer, E. K., & Israel, M. (2024). Artificial intelligence unplugged: designing unplugged activities for a conversational AI summer camp. In *Proceedings of the 55th ACM Technical Symposium on Computing Science Education* (SIGCSE 2024), March 15–18, 2024, Portland, USA. ACM, New York, NY, USA, p. 1272-1278. DOI: https://doi.org/10.1145/3626252.3630783.

14. **Song, Y**., Xing, W., Barron A., Oh, H., Li, C., Minces, V. (2023). M-flow, a flow-based music creation platform improves underrepresented children's attitudes toward computer programming. In *Proceedings of Interaction Design and Children (IDC '23),* June 19–23, 2023, Chicago, IL, USA. ACM, New York, NY, USA, 6 pages. DOI: https://doi.org/ 10.1145/3585088.3589383 **(BEST PAPER)**

15. **Song, Y**., Tian, X., Barrett, J., Israel, M., & Boyer, E. K. (2023). Guide, safety net, project tester, and more: investigating the roles of facilitators in an artificial intelligence summer camp. In *Proceedings of 2023 International Conference of the Learning Sciences* (ICLS 2023). DOI: https://doi.org/10.22318/icls2023.548176.

16. **Song, Y.,** Xing, W., Tian, X., & Li, C. (2023). Are we on the same page? modeling linguistic synchrony and math literacy in mathematical discussions. In *Proceedings of 2023 International Conference on Learning Analytics & Knowledge* (LAK 2023), March 13-17, Arlington, Texas, USA. DOI: doi.acm.org?doi=3576050.3576082 **(BEST PAPER)**

17. **Song, Y**., Katuka, A. G, Barrett, J., Tian, X., Kumar, A., McKlin, T, Celepkolu, M., Israel, M., & Boyer, K. (2023). AI Made By Youth: A conversational AI curriculum for middle school summer camps. In *Proceedings of 2023 Association for the Advancement of Artificial Intelligence* (AAAI 2023), February 7-14, Washington, DC, USA, p. 15851-15859. https://doi.org/10.1609/aaai.v37i13.26882.

18. Katuka, A. G., Auguste, Y., **Song, Y**., Tian, X., Kumar, A., Celepkolu, M., Boyer, K., Barrett, J., Israel M., & McKlin, T. (2023). A Summer camp experience to engage middle school learners in AI through conversational app development. In *Proceedings of the 54th ACM Technical Symposium on Computing Science Education* V. 1 (SIGCSE 2023), March 15–18, 2023, Toronto, ON, Canada. ACM, New York, NY, USA, p. 813-819. DOI: https://doi.org/10.1145/3545945.3569864 **(BEST PAPER)**

EXTENDED ABSTRACT PUBLISHED IN CONFERENCE PROCEEDINGS

1. **Song, Y.,** Ma, Y., Zhu, W., Xing, W. (Mar 2024). *Understanding tutoring strategies of peer and expert tutors in online math discussions using Ordered Network Analysis (ONA).* Poster presented at 2024 Learning Analytics and Knowledge (LAK). Kyoto, Japan.

2. **Song, Y.,** Li, C., Xing, W. (Mar 2024). *An artificial intelligence teachable agent to support learning-by-teaching of secondary school students.* Interactive demo session at 2024 Learning Analytics and Knowledge (LAK). Kyoto, Japan.

3. **Song, Y.,** Xing, W., Li, C. (Mar 2024). *ART-Math: a Story-based Creative Math Learning Platform.* Interactive demo session at 2024 Learning Analytics and Knowledge (LAK). Kyoto, Japan.

BOOKS

- Lim, C., Lee, U., **Song, Y.,** Lee, J., & Choi, S. (2021). *Play with software: data science for elementary students*. Seoul: Naver Connect Foundation. (Online Open Access: https://www.playsw.or.kr/artificial/textbook.)

CONTRIBUTIONS TO EDITED VOLUMES

- **Song, Y.** (2023). Instructional model for data-driven debate class. In Park, J. (Ed.). *Korean Speech Education.* The Korean Association of Speech Communication.

RESEARCH REPORTS SUBMITTED TO SPONSORS

- Lim, C., Ji, H., Park, J., **Song, Y**., & Lee, D. (2020). *An evaluation report on students' satisfaction of educational service* (Korean). Unpublished technical report, Education Research Institute, Seoul National University, Seoul, Korea.
- Lim, C., Lee, U., **Song, Y**., & Jeong, H. (2020). *Development of SMART learning environment for untact era* (Korean). A *research report submitted to Korea Education and Research Information Service (KERIS), Daegu, Korea.*

TEACHING EXPERIENCE

TEACHING IN HIGHER EDUCATION

Instructor of Record

Introduction to Educational Technology — **University of Tennessee, Knoxville**
(ETEC 386) — Knoxville, TN (Online)
Main Instructor, — *Spring, 2026*

AI for Educators — **University of Tennessee, Knoxville**
(Undergraduate and Graduate level, TPTE495, TPTE595) — Knoxville, TN (Online)
Main Instructor, — *Winter 2026*

Teaching Assistant/ Assistant Instructor

Computer Science Pedagogy I — **University of Florida**
(Graduate level, ETE6141) — Gainesville, FL (Online)
Assistant Instructor, — *Fall. 2024*

- Contributed to the course and assignment design and grading.
- Managed the course materials, assignments, and online discussions in the learning management system.
- Designed a cross-walk, cross-talk (CWCT) sessions and gave special lectures on K-12 AI education.

Intro to Quantum Computing — **University of Florida**
(Undergraduate and graduate level, EEE4423/5590) — Gainesville, FL (In-person)
Teaching Assistant, — *Fall. 2024*

- Contributed to the course assignment and learning technology (Spin Qubit).
- Managed the course materials, assignments, and online discussions in the learning management system.
- Graded students' assignments and quizzes.

Instructional Systems Design **Seoul National University**
(Graduate level, 701.652A) Seoul, South Korea (Hybrid)
Assistant Instructor, *Spring. 2020*

- Taught instructional sessions on video and multimedia learning production.
- Arranged and mentored instructional design projects in authentic contexts (e.g., corporate, hospital, school districts).
- Graded and provided feedback on the exams, reflections, and reports.

Software Education for Teaching Subject Matters **Seoul National University**
(Undergraduate level- pre-service teachers, T2184.001900 001) Seoul, South Korea (Hybrid)
Assistant Instructor, *Fall. 2020*

- Taught instructional sessions on integrating data science into subject matter.
- Mentored class projects on designing AI/DS integrated learning activities with different subjects.
- Graded and provided feedback on the exams, presentations, and reports.

Educational Technology **Seoul National University**
(Undergraduate level, 701.308(001)) Seoul, South Korea (Hybrid)
Assistant Instructor, *Fall. 2020*

- Led instructional sessions on creating lesson plans and producing instructional videos.
- Graded and provided feedback on exams, lesson plans, and instructional videos.

Guest lecture

Issues in Trends in Educational Psychology **University of Tennessee, Knoxville**
(Graduate level, EDPY 513) Knoxville, TN (Online)
Guest lecturer, *Nov. 10th, 2025*

- Gave a guest lecture on "Generative AI-powered Personalized Learning Technologies"

Special Topics Seminar on AI in Education **University of Utah**
(Graduate level, ED PS 7960) Salt Lake City, Utah (Online)
Guest lecturer, *Oct. 1. 2024*

- Gave a guest lecture on "Generative AI in K-12 Teaching and Learning."

Doctoral Seminar for Learning Sciences **University of Utah**
(Graduate level, ED PS 7440) Salt Lake City, Utah (Online)
Guest lecturer, *Sep. 30. 2024*

- Gave a guest lecture on "Fair, Accountable, and Transparent (FAccT) Research and Practices: A Fair Clustering."

Seminar in Instructional Technology **University of Alabama**
(Graduate level, AIL 690-910) Tuscaloosa, AL (Online)
Guest lecturer, *Sep. 26. 2024*

- Gave a guest lecture on "Generative AI in Education: Empowering Learning Theories with Emerging Technologies."
- Received an average evaluation score of 4.89 out of 5 on the feedback survey.

Designing Digital Education Curriculum **Xi'an-Jiatong Liverpool University**
(Graduate level, EDS 431) Shanghai, China (Online)
Guest lecturer, *Sep. 21. 2024*

- Gave a guest lecture on "Generative AI in Education: Research and Practices."
- Received an average evaluation score of 4.43 out of 5 on the feedback survey.

K-12 Computer Programming Foundations 1 **University of Florida**
(Graduate level, EDG6805C) Gainesville, FL

Guest lecturer, *Nov. 29. 2023*

- Gave a guest lecture on "Empowering Youth in AI Learning: AMBY interface and DIALOGS curriculum" to CS education graduate students (mostly in-service teachers).

Intro to Multimodal Machine Learning **University of Florida**
(Undergraduate level, CIS 4930) Gainesville, FL
Guest lecturer, *March 23. 2023*

- Gave a guest lecture on "Data Analytics Research and Practices in Education Domain" to CS undergraduate students.
- Received a highly positive qualitative evaluation.

Research on New Media and Program **Korea National University of Transportation**
(Graduate level, 49020) Chungju, Korea
Guest lecturer, "Interdisciplinary learning design for Data Literacy" *Oct. 13, 2021*

- Gave a guest lecture on "Interdisciplinary learning design for Data Literacy" to Ed-tech graduate students (mostly in-service teachers).

TEACHING IN K-12

ART-Math: Creative Math Writing with Generative AI
(P.K.Yonge Developmental Research Elementary school) Gainesville, FL
Instructor, 56 students *May. 2023, Jan-Feb. 2024*

- Instructional design, teaching of 12 5th grade students and 42 6th grade students in formal math classes, learning artifact evaluation.

Learning about Science through AI Chatbot Development Project
(Lincoln Middle School) Gainesville, FL
Instructor, 150 students *Apr. 2024 – June. 2024*

- Instructional design, teaching of 150 6th grade students in formal science classes, learning artifact evaluation.

Camp Dialogs: Middle school AI summer camp, Artificial Intelligence through Conversational AI
(GNV Bridge Community Center) Gainesville, FL
Instructor, 79 students *June. 2022 – August. 2023*

- Iteratively designed and conducted three sessions of a two-week-long in-person summer camp for middle school students.
- Developed 20 lessons and related learning materials including lesson plans, instructional slides, worksheets.

Camp Dialogs Professional Development
(GNV Bridge Community Center, University of Florida) Gainesville, FL
Instructor, 20 undergraduate students *May. 2022 – August. 2023*

- Iteratively designed two years of professional development curricula and lessons.
- Led PD sessions and workshops.

Understanding Artificial Intelligence through Design Thinking
(Afterschool program, 11 public/private high schools) Seoul, South Korea
Instructor, more than 600 students *Jul. 2019 – Jan, 2021*

- Designed the 10-hour after-school program instruction to teach AI through design-thinking project activities.
- Taught a total of more than 600 high school students and gained a 4.48 out of 5 rating.

Creative Math Lab- Introduction to Cryptography
(Afterschool program: Pyeongchang High School) Pyeongchang, South Korea
Instructor, 60 students *Jul. 2020*

- Designed the 4-hour instruction to teach cryptography in a summer camp held in a rural high school.
- Taught in total of 60 high school students and gained a 4.47 out of 5 rating.

Project-based Maker Education
(Summer program, Yeongdeungpo High School) Seoul, South Korea
Facilitator, more than 300 students *Jul. 2019*

- Instructed and facilitated a maker education summer class for around 500 students in a technology-focused high school.

Teaching Internship
(Sookmyung girls' Middle School) Seoul, South Korea
Instructor, 364 students (5.0/5.0; A+) *March – May. 2018*

- Instructed Korean literature and literacy class for 364 middle school students.
- Arranged and conducted individual meetings with 40 middle school students.
- Received an excellent evaluation from in-service teachers (5.0/5.0; A+).

MENTORING

Undergraduate Students
Research Experience for Undergraduates (REUs) *May. 2023–Aug. 2024*

- Mentees: Ms. Carly Solomon, Ms. Madison Edwards, Ms. Nandika Regatti.
- Resulted in publishing lesson packages and conference proceedings.

Graduate Students
Doctoral

- Mentee: Ms. Zhen Zhang, Mr. Ronard Rupard
- Resulted in co-authored papers and presentations in renowned journals and conferences.

PRESENTATIONS

CONFERENCE PRESENTATIONS

1. **Song, Y.,** Kim, J., Moon, J., & Ma, Y. (June, 2026). *Teachers Designing AI: A Mentored Model of Situated Learning in the National AI Challenge.* Presented at the 2026 International Conference on Learning Sciences and Educational Innovation (ICOLSEI). Seoul, South Korea.

2. **Song, Y.,** Ma, Y., Li, C., & Xing, W. (June, 2026). *Reducing Learning Gaps in Learning-by-teaching with GenAI-powered Teachable Agent.* Presented at the 2026 International Conference on Learning Sciences and Educational Innovation (ICOLSEI). Seoul, South Korea.

3. **Song, Y.,** Yang, H., Moon, H., Kilgore, C., & Ma, Y. (June, 2026). *FALCON-AI: Validating an AI Literacy Scale for University Faculty.* Presented at the 2026 International Conference on Learning Sciences and Educational Innovation (ICOLSEI). Seoul, South Korea.

4. Ma, Y. & **Song, Y.** (June, 2026). *Clinical AI Supervision Studio (CAISS): A Case-Based Platform for Training Responsible AI Practice in Clinical Care.* Presented at the 2026 International Conference on Learning Sciences and Educational Innovation (ICOLSEI). Seoul, South Korea.

5. Go, B., Karademir, N., **Song, Y.,** & Lee, K. (July, 2025). *Who is using Artificial Intelligence for What?A Critical Review of Artificial Intelligence in Education Research.* Presented at the 25th International Conference on Educational Research (ICER). Seoul, South Korea.

6. **Song, Y.,** Weisberg, L., Zhang, S., & Israel, M. (April, 2025). *Designing Inclusive AI Learning Experience: A UDL-Guided Framework.* Presented at the 2025 American Educational Research Association (AERA). Denver, Colorado.

7. **Song, Y.,** Li, C., Ma, Y., Lyu, B., Zhu, W., Li, H., & Xing, W. (April, 2025). *What Tutoring Strategies Impact Student' Success and Knowledge Representation in Online Mathematical Discussions?.* Paper session at 2025 American Educational Research Association (AERA). Denver, Colorado.

8. Tian, X., **Song, Y**., Mannekote, A., Israel, M., & Boyer, K. E. (April, 2025). *Automatic evaluation of conversational AI chatbots using large language models.* Presented at 2025 American Educational Research Association (AERA). Denver, Colorado. **(BEST PAPER AWARD NOMINATION)**

9. Tian, X., **Song, Y**., Earle-Randell, T., Zhang, S., Israel, M., & Boyer, K. E. (April, 2025). *"My bot can talk about science!": Fostering AI learning in a middle school science classroom.* Presented at 2025 American Educational Research Association (AERA). Denver, Colorado. **(BEST PAPER AWARD NOMINATION)**

10. Xing, W., **Song, Y**., Li, C., & Liu, Z. (April, 2025). *Flipping the Roles: Engaging Middle School Students in Learning-by-teaching with a Generative AI-powered Teachable Agent.* Paper session at 2025 American Educational Research Association (AERA). Denver, Colorado.

11. Ma, Y., & **Song, Y.** (Apr. 2025). *A Framework Defining AI Literacy for Medical Students.* Presented at 2025 American Educational Research Association (AERA). Denver, Colorado.

12. Liu, Z., **Song, Y.,** & Xing, W. (Apr. 2025). *Identifying factors of in-video dropout in middle school online math learning using survival analysis.* Paper session presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.

13. Xing, W., Liu, Z., & **Song, Y**. (Apr. 2025). *Teaching quantum computing to undergraduate students using a multimedia and simulation-based learning technology.* Roundtable session presented at the 2025 American Educational Research Association (AERA) Annual Meeting, Denver, Colorado.

14. **Song, Y**., Li, C., Ma, Y., Zhu, W., Lyu, B., Li, H., Xing, W. (Oct. 2024). *The Impact of Tutoring Strategies on Success and Knowledge Representation in Asynchronous Online Math Discussions.* Presented at 2024 Association for Educational Communications and Technology (AECT). Kansas City, MO, USA.

15. Li, C., Xing, W., **Song, Y.**, Lyu, B. (Oct. 2024). *A Design-based Research Study on Conversational Artificial Intelligence to Support Math Learning.* Presented at 2024 Association for Educational Communications and Technology (AECT). Kansas City, MO, USA.

16. Liu, Z., **Song, Y.**, Xing, W., Guo, J., Feng, P., & Kim, G. (Oct. 2024). *Enhancing students' quantum computing learning through multimedia and simulation-based tools*. Presented at 2024 Association for Educational Communications and Technology (AECT). Kansas City, MO, USA.

17. Xing, W., Liu, Z., & **Song, Y.** (Oct. 2024). *Understanding in-video dropout behavior: Analyzing student engagement in online math education.* Presented at 2024 Association for Educational Communications and Technology (AECT). Kansas City, MO, USA.

18. Oh, H., Guo, R., Xing, W., **Song, Y.,** Li, C., Liu, Z. (Oct. 2024). *Integrating Machine Learning into High School Mathematics*. Presented at 2024 Association for Educational Communications and Technology (AECT). Kansas City, MO, USA.

19. Yang, Q., William, R., **Song, Y**., Xing, W., Guo, J. (Sep. 2024). *An Interactive Tool of Spin Qubit for Quantum Science and Engineering Education.* Presented at 2024 IEEE International Conference on Quantum Computing and Engineering (QCE24). Montréal, Québec, Canada.

20. **Song, Y**., Li, C., Xing, W., Lee, H., Li, S., Ma, Y. (Apr. 2024). *Fairness-Aware Behavioral Clustering for Self-regulated Learning in Virtual Learning Environments.* Presented at 2024 American Educational Research Association (AERA). Philadelphia, PA, USA.

21. **Song, Y**., Xing, W., Li, C., Oh, H. (Apr. 2024). *ART-Math: A Story-based Math Creation Platform for Engaging Elementary School Learners in Mathematics Learning.* Presented at 2024 American Educational Research Association (AERA). Philadelphia, PA, USA.

22. **Song, Y**., Tian, X., Solomon, C., Israel, M. (Oct 2023). *Demystifying AI for Middle School Learners through Building Chatbots.* AI. AI days, University of Florida.

23. **Song, Y**., Zhang, Z., Oh, H., Li, C., Xing, W. (Oct 2023). *Draw, Tell, Create! A Story-based Mathematics Creation Platform for Elementary School Learners.* 2023 Florida Association for Teacher Educators (FATE).

24. Moore, R. **Song, Y**., Abramowitz, B., & Opoky, R. (Oct 2023). *Let's Build Together: Techniques and Approaches for Engaging Stakeholders in AI Development*. Graduate Student Panel Talk, 2023 Empowering Learners in AI conference, Online.

25. **Song, Y**., Xing, W., Barron, A., Oh, H., Li, C. & Minces, V. (Oct 2023). *Unlocking the Potential of Coding Music for Young Students through M-flow Platform.* Presented at 2023 Association for Educational Communications and Technology (AECT). Orlando, FL, USA.
26. **Song, Y**., Tian, X., Barrett, J., Israel, M., & Boyer, E. K. (11 June 2023). *Guide, Safety Net, Project Tester, and More: Investigating the Roles of Facilitators in an Artificial Intelligence Summer Camp.* Presented at 2023 International Conference of the Learning Sciences (ICLS). Montreal, Canada.

27. **Song, Y**., Ma, Y., & Israel, M. (4 Apr 2023). *Understanding Collaborative Learning in Pair Programming using Ordered Network Analysis.* Presented at College of Education's Research Symposium, University of Florida. **(BEST PAPER AWARD)**

28. **Song, Y**., Myers, T., & Rahimi, S. (Nov 2022). *Stealth Assessment Theory, design, and practice: Computational Thinking*. Presented at Florida Educational Research Association (FERA). Daytona Beach, FL.

29. Lim, C., **Song, Y**., Lim, E., & Lee, J. (28 Oct 2022). *A Study on Development and Application of a Data Science Education Program for Elementary School Students.* Presented at 2022 Association for Educational Communications and Technology (AECT).

30. Tian, X., **Song, Y.,** & Kumar, A. (17 May 2022). *Teaching Artificial Intelligence in Middle School Summer Camps through Conversational AI : Development of DIALOGS Curriculum and AMBY Interface.* Presented at AI in Education Symposium, University of Florida.

31. Myers, T. & **Song, Y.** (14 Apr 2022). *ACTS: Assessing Computational Thinking Stealthily in a Block Coding-based Game Environment.* Presented at College of Education's Research Symposium, University of Florida. **(HONORABLE MENTION POSTER AWARD)**

32. **Song, Y.** (2020). *Instructional Model for Design Thinking-based AI Service Designing Program for high school students.* Presented at 2020 Association for Educational Communications and Technology (AECT).

33. Lim, C., Han, H., Hong, S., **Song, Y.**, & Lee, D. (2020). *Design Principles for Improving Creative Thinking Competency in Corporate* Education. Presented at 2020 Association for Educational Communications and Technology (AECT).

34. Lim, C., Lee, U., Bae, Y., **Song, Y**., Lee, J., & Jung, H. (2020). *Smart Learning Environment for the Post-COVID era.* Korean Association for Educational Information and Media & the Korean Society for Educational Technology.

35. **Song, Y.,** Hong, S., Park, C., Liu, K., Lim, C. (2019). *A study on the improvement of RPISD (rapid prototyping to ISD).* Presented at 2019 International Conference on Education Research (ICER).

36. Lim, C., Han, H., Lee, D., **Song, Y.,** Hong, S. (2019*). Development and Implementation of Educational Methodology for creative thinking in corporate contexts.* Presented at Korean Association for Educational Information and Media & the Korean Society for Educational Technology.

37. **Song, Y.** (2019). *Development and Implementation of the Design Thinking-Based Software Planning Program- in high school context.* Presented at 2019 International Conference on Education Research (ICER).

38. **Song, Y.**, Hong, S., Liu, K., Park, C. (2019). *A Study on the Development of Creative Marketing Educational Training Program based on RPISD (Rapid Prototyping to ISD).* Presented at 2019 International Conference on Education Research (ICER).

WORKSHOPS

- Grover, S. **Song, Y**., et al., (2023, April). *Learning AI through Youth-led Conversational App Development Experiences.* Presented at Teaching Artificial Intelligence in K-12: Innovative Learning Designs in an Evolving Landscape workshop, American Educational Research Association (AERA).
- Israel, M., Barrett, J., McKlin, **Song, Y.,** Tian, X. (2023, February). *Curricular Design Decisions for Camp DIALOGS: An Inclusive AI Summer Camp Experience for Middle Schoolers*. Presented at Inclusive and Diverse K-12 AI Literacy workshop, 2023 Association for the Advancement of Artificial Intelligence (AAAI).

INVITED TALKS

1. "Towards Inclusive AIED" Research meeting for graduate research assistants, AI Across the Curriculum for Virtual School, Education Innovation and Research (EIR) Program, Concord Consortium *(Oct, 2024).*
2. "Learning about AI by Creating Conversational Ais: Summer Camp and Classroom Experiences." Professional Development for middle school teachers, Integrating AI Machine Learning into Teaching of Paleontology Using Fossil Shark Teeth in Middle Schoolers, NSF ITEST (#2147625), University of Florida *(June, 2024).*
3. "How can we make AI learning more inclusive: a new framework," CAST AI Quarterly Convening, Center for Applied Special Technology (CAST) *(June, 2024).*
4. "Camp Dialogs: Teaching AI through Conversational AI," Professional Development for middle school teachers, Integrating AI Machine Learning into Teaching of Paleontology Using Fossil Shark Teeth in Middle Schoolers, NSF ITEST (#2147625), University of Florida *(July, 2023)*
5. "When Education Meets AI/Data Science: Research in Learning Analytics" Educational Reading Group (a study/research group of 20~30 computer science and education Ph.D. students), Stanford University *(May 8, 2023)*
6. "AI Education, Maker Education," discussion panel at Korean EduTech and Learning Science (KELS), online *(February, 2023)*
7. "Data Science-integrated education for Data Literacy," AI Education Symposium, Department of AI-Integrated Education, Seoul National University, South Korea (*January, 2023*)
8. "Leveraging Learning Analytics to Create New Edu Tech Service," National IT Industry Promotion Agency, South Korea *(August, 2022)*
9. "Teaching AI through Conversational AI," Professional Development for middle school teachers, Integrating AI Machine Learning into Teaching of Paleontology Using Fossil Shark Teeth in Middle Schoolers, NSF ITEST (#2147625), University of Florida *(July, 2022)*
10. "A Step for the Future Education," Korean-American Scientists and Engineers Association (KSEA) Spring Gainesville Chapter, University of Florida *(Apr 2, 2022).*
11. "How to integrate data science into classrooms," Professional Development, Gyeonggido Suwon Office of Education *(Oct, 2021)*
12. "Teaching AI/Data Literacy in classrooms," Professional Development, Nanwoo Middle School *(May ~ Sep, 2021)*

LEADERSHIP & SERVICE

DEPARTMENT/ ACADEMIC EVENTS

Student Alliance of Graduates in Education (SAGE) — **University of Florida**
President (serving for graduate students' academic, professional and social lives) — *Fall. 2024 – Summer. 2025*

College of Education Research Symposium Planning Committee — **University of Florida**
Engagement chair (recruit faculty reviewers and selected awardees) — *Jan. 2024 – March. 2024*

Program Search Committee — **University of Florida**
Graduate student volunteer (candidate review and event coordination) — *Jan. 2024*

Association for Educational Communication and Technology (AECT) **Orlando, FL**
Session chair (session facilitation) "Understanding Learners and Learning Environments" *Oct. 2023*

AI Days Event **University of Florida**
Organizing committee (organized tabling) *Oct. 2023*

AI in Education Workshop **University of Florida**
Session facilitator (online) "AI in Assessment and Improving Learning" *Apr. 2023*

Computer Science Education Week (CS Ed Week) **University of Florida**
Organizing committee (organized tabling) *Dec. 2022*

International Conference on Educational Research (ICER) **Seoul National University**
Organizing committee (international outreach) *Oct. 2019*

CONSULTING

Korea University, Sejong **Republic of Korea**
Consultant, What core curriculum should universities establish in the age of AI *Feb. – July. 2026*

- AI core curriculum development for higher education in Korean context

Korean Department of Education **Republic of Korea**
Consultant, AI Education Alliance and Policy Lab (AIEDAP) project *Oct. 2023*

- Provided a consultation regarding the integration of AI into subject matters.

Seoul Metropolitan Office of Education **Republic of Korea**
Consultant, AI/Data Literacy School Project *Sep. 2020 – Dec. 2021*

- Provided a consultation on the integration of AI/Data Science into subject matters.

JOURNAL REVIEW

- Journal of Computer Assisted Learning (2026)
- Computers and Education (C&E) (2024-Present)
- Journal of Research on Technology in Education (JRTE) (2024-Present)
- The Internet and Higher Education (2024)
- Journal of Science Education and Technology (2024-Present)
- Journal of Educational Computing Research (JECR) (2024-Present)
- International Journal of Artificial Intelligence in Education (IJAIED) (2023-Present)

CONFERENCE REVIEW

- Interaction Design and Children (IDC) (2024)
- Special Interest Group for Computer Science Education (SIGCSE) (2024)
- Association for Educational Communications and Technology (AECT) (2023-Present)
- International Society of the Learning Sciences (ISLS) (2022)
- American Educational Research Association (AERA) (2022-Present)

OTHER SERVICE

- Academic and Professional Mentoring Service, Korean-American Educational Researchers Association (KAERA) (2026)
- Instructor/ Director of Academic Affairs, Gainesville Korean School (2022 – 2025)
- Vice president at the Korean Student Tennis Club, the University of Florida (2022)
- A media manager for an animal shelter located in South Korea, e.g., created social media postings to improve public perception about dog-meat and animal abuses (2021)

- Teaching English to underprivileged students in South Korea, Seoul Urban Science and Technical High School and Donggu Marketing High School (2017)
- Teaching Korean at the Korean Student Association, the University of North Carolina at Greensboro (2016)
- Mural painting in a countryside village in South Korea (2015)

PROFESSIONAL AFFILIATIONS

- ACM Interaction Design and Children (IDC) *(2022-2023)*
- Learning Analytics and Knowledge (LAK) *(2022-Present)*
- International Society of Learning Sciences (ISLS) *(2022-2023)*
- The Association for the Advancement of Artificial Intelligence (AAAI) *(2022-2023)*
- Special Interest Group in Computer Science Education (SIGCSE) *(2022-2025)*
- American Educational Research Association (AERA) *(2022-Present)*
- Association for Educational Communication and Technology (AECT) *(2019-Present)*
- Korean Society of Educational Technology (KSET) *(2019-2021)*

CERTIFICATES

- Teacher's License for secondary education (by the Ministry of Education, Science and Technology, Rep. of Korea)
- Certificate of Completion of the Preparing Future Faculty Program, Center for Teaching Excellence at the University of Florida.
- Tensorflow Developer Certificate (by Google)
- Certificate of Korean Language Teacher (by Ministry of Culture, Sports and Tourism, Rep. of Korea)
- Certificate of Korean History Lv. 3 (by National Institute of Korean History, Rep. of Korea)
- Human Resources Development Instructor Certificate (by KHRDA: Korean Human Resources Development Association)

LANGUAGES AND SKILLS

- Korean (Native)
- English (Fluent)
- Chinese (Conversational): HSK Lv. 4 (265)
- Python, R, HTML

PATENTS

- Method for controlling car seat position-based learning (Korean patent, Under Review, #1020210160162)

REFERENCES

Dr. Wanli Xing – University of Florida
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Dr. Maya Israel – University of Florida
Associate Professor of Educational Technology and Computer Science Education
Director of CSEveryone: The Kenneth C. Griffin CS for All Education initiative
Director of Creative Technology Research Lab
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Dr. Kristy Boyer – University of Florida
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Director of LearnDialogue Lab
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Dr. Chenglu Li – The University of Utah
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